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RESEARCH ARTICLE

An Explainable Machine Learning Framework for Organizational Financial Distress Detection: Evidence From Korean Hospital Panel Data

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ABSTRACT Detecting organizational financial distress is a fundamental problem in machine learning for finance, yet existing approaches predominantly target corporate bankruptcy using stock market data, leaving non-market organizations—such as hospitals, universities, and public institutions—underexplored. We present a systematic explainable machine learning framework comprising four stages: domain-adaptive feature engineering, multi-model training with ensemble methods, SHAP-based global and local interpretation, and domain-specific validation. We instantiate this framework on Korean hospital panel data, constructing a balanced panel of 165 hospitals (825 hospital-year observations, 2019–2023) from mandatory public financial disclosures containing 236 line items per observation. We engineer 11 features capturing cost structure, financial position, institutional characteristics, and external subsidy effects. Among ensemble classifiers evaluated under hospital-level Group K-Fold cross-validation (preventing within-entity data leakage), XGBoost achieves the highest AUC of 0.842, closely followed by Gradient Boosting (0.840) and Random Forest (0.831), all consistently outperforming Logistic Regression (0.776). Temporal validation (training on 2019–2022, testing on 2023) yields AUC = 0.780. Note that tree-based ensembles cannot extrapolate beyond their training domain for continuous variables. Because the ‘year’ feature values in the test set (‘year=4’) exceed the training distribution (‘year’ 0–3), the models evaluate 2023 data using the terminal-node logic for ‘year=3’. The AUC of 0.780 therefore reflects the robustness of the cost-structure features in detecting distress even without explicit temporal trend extrapolation. SHAP analysis reveals that material cost ratio (mean |SHAP| = 0.777) and labor cost ratio (0.729) are the dominant predictive features, while external subsidies—despite strong directional protective effects—rank low in overall importance due to limited variance. We further demonstrate the framework’s utility for counterfactual analysis by constructing subsidy-adjusted distress trajectories that reveal a 57.5 percentage point gap between observed and latent distress rates, and for panel econometric validation via fixed effects regression, which shows that the post-subsidy period increased distress probability by 28.9 percentage points ($p < 0.001$). The approach is designed to be adaptable to other organizational settings with structured financial disclosures and is publicly available.

INDEX TERMS Explainable artificial intelligence, financial distress prediction, SHAP, gradient boosting, feature engineering, panel data, hospital finance, interpretable machine learning.

I. INTRODUCTION

Financial distress prediction is a long-standing challenge at the intersection of machine learning and finance. Since

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Altman’s [1] seminal Z-score model and Ohlson’s [2] logistic regression approach, the field has evolved from linear discriminant models to sophisticated ensemble methods including Random Forests [3], Gradient Boosting [4], and XGBoost [5], achieving progressively higher predictive accuracy [6], [7], [8]. However, two critical gaps persist.

First, the vast majority of financial distress prediction research focuses on publicly traded corporations, using stock market signals as leading indicators [6], [9]. Non-market organizations—hospitals, universities, government agencies, and other entities that do not trade on public exchanges—represent a substantial and underexplored class of financial distress prediction targets. These organizations often operate under regulated revenue constraints, carry public service mandates, and face cost structures fundamentally different from those of for-profit corporations. Existing corporate bankruptcy models transfer poorly to such settings because the distress mechanisms differ: rather than equity depletion or debt default, non-market organizations typically experience progressive margin compression and operating deficits that may persist for years without triggering formal insolvency [10].

Second, the push toward explainable AI (XAI) has highlighted that predictive accuracy alone is insufficient for high-stakes financial applications [11], [12]. Stakeholders—regulators, managers, policymakers—require not only predictions but also actionable explanations of *why* an organization is predicted to be in distress and *which* factors are most amenable to intervention [13], [14]. While SHAP (SHapley Additive exPlanations) [15], [16] is now recognized as the leading post-hoc explanation method for tree-based models, systematic frameworks that integrate SHAP-based interpretation into end-to-end distress prediction pipelines remain scarce [17].

To address these gaps, we present a **systematic explainable machine learning framework for organizational financial distress detection** comprising four stages: (1) domain-adaptive feature engineering from structured financial disclosures, (2) multi-model comparison across linear, ensemble, and gradient-boosted classifiers, (3) SHAP-based global and local interpretation with compositional awareness, and (4) domain-specific validation including entity-level cross-validation, temporal holdout, and counterfactual analysis.

We instantiate the framework on a challenging real-world application: detecting financial distress among Korean hospitals using mandatory public financial disclosure data. The hospital domain is particularly well-suited for validating an organizational distress detection framework for several reasons. First, Korean law requires hospitals above a size threshold to publicly disclose audited financial statements through the Korea Health Industry Development Institute (KHIDI) portal, providing a rich panel dataset with 236 financial line items per hospital-year—far more granular than typical corporate datasets [18]. Second, the hospital sector experienced a natural experiment during 2019–2023: the COVID-19 pandemic triggered massive government subsidies (2020–2022) followed by abrupt withdrawal (2023), creating an external shock that shifted the distress rate from 35.2% to 69.1%—an ideal testbed for evaluating model robustness under distributional shift. Third, hospital financial

distress has direct societal consequences affecting health-care access, making interpretable predictions actionable for policy [19], [20].

Our contributions are as follows:

- 1) **Framework design.** We present a four-stage explainable ML framework for organizational financial distress detection. Although it assembles standard components (data preprocessing, XGBoost training, SHAP analysis, and K-Fold validation), the framework's primary contribution lies in domain-adaptive integration: mapping complex accounting identities into ML features, adjusting for compositional collinearity, and using SHAP to quantify the exact bounds of subsidy masking effects in non-market organizations.
- 2) **Feature engineering methodology.** We develop a principled feature engineering pipeline that transforms 236 raw financial line items into 11 interpretable features spanning cost structure ratios, financial position indicators, institutional categorical variables, and temporal trend features. We demonstrate how domain knowledge guides feature selection and analyze potential endogeneity in subsidy-related features.
- 3) **Model comparison and evaluation.** We systematically compare four classifiers—Gradient Boosting, XGBoost, Random Forest, and Logistic Regression—under three evaluation protocols: hospital-level Group K-Fold (primary), temporal validation (robustness), and standard stratified K-Fold (reference). XGBoost achieves $AUC = 0.842$ under the strictest protocol.
- 4) **SHAP-based distress interpretation.** We provide a detailed SHAP analysis revealing that cost structure ratios (material: mean $|SHAP| = 0.777$; labor: 0.729) dominate distress prediction, while external subsidies—despite strong local protective effects—contribute limited global importance due to low variance. We reconcile the divergence between SHAP rankings and logistic regression coefficients through a variance-vs-effect size analysis.
- 5) **Counterfactual and econometric validation.** We demonstrate the framework's utility for counterfactual distress analysis by constructing subsidy-adjusted trajectories that reveal latent distress masked by temporary external transfers. Panel fixed effects regression confirms the post-subsidy structural break (28.9 pp, $p < 0.001$), providing econometric triangulation of the ML findings.

The remainder of this paper is organized as follows. Section II reviews related work on financial distress prediction and explainable AI. Section III presents the explainable ML framework architecture. Section IV describes the experimental setup, including dataset, features, and evaluation protocols. Section V presents experimental results. Section VI discusses technical insights and framework generalizability. Section VII concludes.

II. RELATED WORK

A. FINANCIAL DISTRESS PREDICTION WITH MACHINE LEARNING

Financial distress prediction has evolved through three generations of methods. The first generation established linear models: Altman's [1] Z-score used five financial ratios in a linear discriminant function, and Ohlson [2] introduced logistic regression to relax distributional assumptions. Altman later proposed the Z''-score [21] for non-manufacturing and private firms, removing the market equity component that limited applicability to non-traded organizations. These models demonstrated that financial ratios—profitability, leverage, liquidity—contain substantial predictive information about organizational viability.

The second generation introduced ensemble methods. Breiman's [3] Random Forest aggregated decision trees over bootstrap samples, reducing overfitting while capturing nonlinear relationships. Friedman's [4] Gradient Boosting Machine (GBM) sequentially corrected prediction errors through additive tree ensembles. Chen and Guestrin's [5] XGBoost added regularization and scalable parallelization, becoming a dominant tool in financial prediction tasks. Barboza et al. [6] provided an extensive comparison showing that ensemble methods outperform traditional statistical models for bankruptcy prediction. Zięba et al. [7] demonstrated that ensemble boosted trees with synthetic feature generation further improve predictive performance.

The third generation incorporates deep learning. Mai et al. [9] applied recurrent neural networks to textual disclosures for bankruptcy prediction. Moscatelli et al. [8] combined traditional financial features with machine learning for corporate default forecasting. However, deep learning approaches sacrifice interpretability—a critical requirement in regulated financial domains—and typically require larger training datasets than are available in most organizational distress settings.

A key limitation across all three generations is their near-exclusive focus on publicly traded corporations. Non-market organizations—hospitals [10], municipalities, universities—face fundamentally different distress dynamics: regulated revenue, public service obligations, soft budget constraints [22], and prolonged tolerance of operating deficits. Our work extends financial distress prediction to this underexplored organizational class.

B. EXPLAINABLE AI FOR FINANCIAL APPLICATIONS

The need for model interpretability in financial applications has driven significant recent development in XAI methods [11]. Ribeiro et al. [23] introduced LIME (Local Interpretable Model-agnostic Explanations), which approximates complex models locally with interpretable surrogates. Lundberg and Lee [15] proposed SHAP, grounding feature attribution in Shapley values from cooperative game theory [24], providing uniquely fair and complete decompositions of model predictions. Lundberg et al. [16] extended

SHAP with TreeExplainer, enabling exact and efficient computation for tree-based models and demonstrating its utility in healthcare applications.

Several frameworks have been proposed for integrating XAI into financial pipelines. Černevičienė and Kabašinskas [17] reviewed explainable AI in finance, identified SHAP as the most widely adopted method, and noted that few studies provide systematic frameworks for end-to-end explainable prediction. Du et al. [13] surveyed interpretable ML techniques and identified the gap between model-level interpretability and decision-level actionability. Molnar et al. [14] formalized the distinction between inherently interpretable models (e.g., logistic regression) and post-hoc explanation methods (e.g., SHAP), arguing that both are needed for complete understanding. Doshi-Velez and Kim [12] called for rigorous evaluation of interpretability claims, noting that SHAP values alone do not guarantee actionable insights without domain-contextual analysis.

More recent studies have applied explainable ML specifically to financial-risk prediction: Bussmann et al. [25] use Shapley values to explain credit-risk models for SME lending, and Gramegna and Giudici [26] evaluate the discriminative power of SHAP and LIME for credit risk. These works underscore the growing role of post-hoc explanation in financial decision-making, which our framework extends to the under-studied setting of non-market organizations.

Our work contributes to this literature by demonstrating how SHAP analysis can be combined with domain knowledge to yield actionable distress explanations—not merely feature rankings but mechanistic insights that inform intervention strategies.

C. PANEL DATA CHALLENGES IN ML EVALUATION

A key methodological challenge in applying ML to panel data is the risk of within-entity data leakage. When multiple observations from the same entity (e.g., hospital, firm) appear in both the training and test sets, the model may exploit entity-specific patterns rather than learning generalizable predictive features. Hastie et al. [27] discuss cross-validation principles but do not specifically address panel data structures. We address this challenge through Group K-Fold cross-validation [28], where entire entities are held out per fold, providing unbiased performance estimates on unseen organizations. We additionally employ temporal validation (training on earlier years and testing on later years) to evaluate robustness under distributional shift—a critical consideration given that distress rates may change over time due to exogenous shocks.

III. EXPLAINABLE ML FRAMEWORK

We present a four-stage framework for organizational financial distress detection, designed to be domain-agnostic in architecture while domain-adaptive in implementation. The framework architecture is described below.

A. STAGE 1: DOMAIN-ADAPTIVE FEATURE ENGINEERING

The first stage transforms raw financial disclosures into a compact, interpretable feature set through a principled process:

- 1) **Raw data ingestion.** Structured financial statements (balance sheets, income statements) are parsed from disclosure portals. For the hospital domain, this yields 236 line items per entity-year from the KHIDI/HASPA portal.
- 2) **Ratio construction.** Raw monetary values are converted into interpretable ratios that normalize for entity size and enable cross-entity comparison. Cost structure ratios (e.g., labor cost / revenue, material cost / revenue) are particularly informative as they capture the operational efficiency that determines distress vulnerability.
- 3) **Categorical encoding.** Domain-specific institutional characteristics (e.g., entity type, governance structure, tier designation) are encoded as binary indicators. These features capture structural cost burdens not reflected in financial ratios.
- 4) **Temporal and scale features.** Time indices capture secular trends, while log-transformed size indicators (e.g., revenue, capacity) account for scale effects. The logarithmic transformation compresses skewed distributions and prevents extreme values from dominating.
- 5) **Preprocessing.** Missing values are handled via listwise deletion when rare ($< 1\%$); ratios are winsorized at the 1st and 99th percentiles to limit outlier influence while preserving distributional shape.

The key design principle is that features should be (a) interpretable to domain stakeholders, (b) computable from publicly available disclosures, and (c) free from look-ahead bias. The reduction from 236 raw line items to 11 features follows a systematic three-step process: (1) domain expert consultation identified five conceptual categories (cost structure, revenue composition, financial position, institutional type, and temporal effects); (2) within each category, raw line items were converted to ratios to ensure scale-invariance across hospital sizes; (3) highly correlated features ($r > 0.85$) were removed to minimize multicollinearity (e.g., total revenue and total cost were replaced by their ratio decomposition). This principled selection ensures interpretability while avoiding the curse of dimensionality on a modestly-sized dataset ($n = 825$). Domain knowledge also guides the identification of potentially endogenous features (e.g., subsidy ratios that have near-mechanical relationships with the target).

B. STAGE 2: MULTI-MODEL TRAINING

The second stage trains multiple classifiers spanning a range of model complexity:

- **Linear baseline** (Logistic Regression [29]): Provides interpretable coefficients and serves as a performance

floor. L2 regularization prevents overfitting; features are standardized to zero mean and unit variance.

- **Bagging ensemble** (Random Forest [3]): Aggregates predictions across bootstrap-trained decision trees with random feature subsets, capturing nonlinear relationships without explicit specification.
- **Boosting ensembles** (Gradient Boosting [4], XGBoost [5]): Sequentially build shallow trees correcting prior residuals. The scikit-learn GBM implementation [30] uses 100 trees (max depth 3, learning rate 0.1); XGBoost adds L1/L2 regularization.

Hyperparameters are set to standard defaults rather than tuned via grid search, reducing overfitting risk and enhancing reproducibility. Class rebalancing is not applied when the minority class exceeds 30%.

C. STAGE 3: SHAP-BASED INTERPRETATION

The third stage applies SHAP [15] to the best-performing model to produce:

- **Global feature importance:** Mean absolute SHAP values across all observations, providing a ranking of feature contributions to distress prediction.
- **Directional effects:** SHAP beeswarm plots revealing both the magnitude and direction of each feature's impact, distinguishing risk factors (positive SHAP for high values) from protective factors (negative SHAP for high values).
- **Compositional awareness:** When features represent compositional shares (e.g., cost ratios summing to a total), SHAP values for individual components may be unstable due to collinearity. The framework flags this and reports aggregate importance for feature groups.
- **Cross-method reconciliation:** When SHAP rankings diverge from linear model coefficients, the framework provides a variance-vs-effect size analysis explaining the divergence (features with strong effects but low variance may rank high in linear models but low in SHAP).

SHAP values are computed using TreeExplainer [16], which provides exact Shapley values for tree-based models in polynomial time.

D. STAGE 4: DOMAIN-SPECIFIC VALIDATION

The fourth stage validates model performance and interpretability through multiple protocols:

- **Entity-level Group K-Fold:** The primary evaluation protocol, where entire entities are held out per fold to prevent within-entity data leakage. This provides unbiased estimates of performance on unseen organizations.
- **Temporal validation:** Training on earlier periods and testing on later periods evaluates robustness under distributional shift (e.g., regime changes, external shocks).
- **Counterfactual analysis:** Domain-specific scenario construction (e.g., removing external subsidies, adjusting cost growth) tests model-implied distress rates

TABLE 1. Panel composition summary.

Category	Count	%
<i>Hospital Type</i>		
Tertiary referral	27	16.4
General	138	83.6
<i>Establishment Type</i>		
Medical corporation	74	44.8
Public	42	25.5
University	35	21.2
Other	14	8.5
<i>Panel Dimensions</i>		
Entities (hospitals)	165	—
Time periods (years)	5	—
Total observations	825	—
Features per observation	236	—

Data source: KHIDI/HASPA hospital financial disclosure (2019–2023). The panel represents the near-census of Korean tertiary and general hospitals reporting continuously over the study period.

under alternative assumptions, providing policy-relevant insights.

- **Econometric triangulation:** Panel fixed effects regression provides complementary causal evidence, confirming that the features identified by SHAP as important are also statistically significant predictors under standard econometric identification strategies.

IV. EXPERIMENTAL SETUP

A. DATASET DESCRIPTION

We instantiate the framework on Korean hospital financial disclosure data from the KHIDI portal. Under Article 62 of the Korean Medical Service Act, hospitals with 300 or more beds must publicly disclose audited financial statements annually, comprising 236 distinct financial line items (balance sheet and income statement). We construct a balanced panel of 165 hospitals observed continuously from 2019 to 2023, yielding 825 hospital-year observations. Table 1 summarizes the panel composition.

The balanced panel design ensures that all temporal comparisons reflect the same entities, eliminating compositional bias from entry/exit. We adopt a superpopulation inference framework [31], treating the observed data as one realization from a stochastic data-generating process, which justifies hypothesis testing despite near-complete population coverage.

1) PREDICTION TARGET

The binary deficit indicator serves as the prediction target: $Y_{it} = 1$ if net income < 0 for hospital i in year t , and $Y_{it} = 0$ otherwise. The deficit rate varies substantially across time: 35.2% (2019), 46.7% (2020), 26.1% (2021), 29.7% (2022), and 69.1% (2023), reflecting a natural experiment created by pandemic subsidy injection and withdrawal. The overall class balance (41.2% positive) does not require rebalancing.

TABLE 2. Financial trajectory of 165 Korean hospitals (2019–2023).

Metric	2019	2020	2021	2022	2023
Deficit rate (%)	35.2	46.7	26.1	29.7	69.1
Med. rev. (mean, $\times 100$ M KRW)	1,410	1,376	1,493	1,604	1,736
Med. cost (mean, $\times 100$ M KRW)	1,384	1,401	1,512	1,644	1,795
Op. margin (%)	+1.83	−1.82	−1.27	−2.49	−3.39

Monetary values in hundred million KRW ($\times 100$ M KRW). McNemar test (2019 vs. 2023): $p = 4.91 \times 10^{-11}$. All Wilcoxon signed-rank tests: $p < 0.001$.

TABLE 3. Cost decomposition: Aggregate increase 2019–2023 (165 hospitals).

Cost Category	Increase (trillion KRW)	Share (%)
Labor costs	3.259	48.1
Material costs	2.365	34.9
Administrative costs	1.147	16.9
Total	6.773	100.0

Amounts are in trillions of KRW. Labor/revenue ratio: 45.7% (2019) $\rightarrow$ 48.5% (2023).

TABLE 4. Engineered feature set for distress prediction.

Feature	Description
<i>Cost Structure Ratios</i>	
material_ratio	Material costs / Medical revenue
labor_ratio	Labor costs / Medical revenue
admin_ratio	Administrative costs / Medical revenue
<i>Financial Position</i>	
subsidy_ratio	Donation revenue / Total revenue
debt_ratio	Total liabilities / Total assets
<i>Scale and Temporal</i>	
log_revenue	$\ln(\text{Medical revenue})$
log_beds	$\ln(\text{Number of beds})$
year	Integer (2019=0, ..., 2023=4)
<i>Institutional Categories</i>	
is_university	University-affiliated hospital
is_public	Publicly operated hospital
is_tertiary	Tertiary referral designation

Features are pre-specified based on domain knowledge. Cost structure ratios are compositionally related (shares of total cost). The year and subsidy_ratio features require careful interpretation due to potential endogeneity (see text).

2) DOMAIN CONTEXT: REVENUE–COST DYNAMICS

Between 2019 and 2023, hospital mean revenue grew 23.1% while mean cost grew 29.7%, producing a “scissors effect” that compressed the medical operating margin from +1.83% to −3.39% (Table 2). The total cost increase across the panel was 6.773 trillion KRW, with labor costs accounting for 48.1%, material costs for 34.9%, and administrative costs for 16.9% (Table 3). Government COVID-19 subsidies (recorded as donation revenue) peaked at 2.159 trillion KRW in 2021 before declining to 0.530 trillion KRW in 2023. This creates a challenging prediction setting in which the distress-generating mechanism shifts over time.

B. FEATURE ENGINEERING

Following Stage 1 of the framework (Section III-A), we engineer 11 features from the 236 raw financial line items. Table 4 summarizes the feature set.

Two features warrant specific discussion regarding potential endogeneity. The `year` feature captures secular trends in distress rates but may proxy for unobserved time-varying confounders; model performance excluding this feature should be examined. The `subsidy_ratio` has a near-mechanical relationship with the deficit outcome, since higher subsidy ratios directly increase net income. We retain both features in the primary analysis for completeness but flag them as robustness considerations.

The three cost ratio features (labor, material, administrative) are compositionally constrained as shares of total medical cost. This collinearity can produce instability in individual SHAP attributions; we address this by reporting aggregate cost-ratio importance alongside the individual rankings.

C. MODEL SPECIFICATIONS

Following Stage 2 (Section III-B), we train five classifiers:

- 1) **Logistic Regression** [29]: L2 regularization ($C = 1.0$), standardized features. Provides an interpretable coefficient baseline.
- 2) **Random Forest** [3]: 100 trees, bootstrap sampling, random feature subsets per split.
- 3) **Gradient Boosting** [4]: 100 trees, max depth 3, learning rate 0.1. Scikit-learn [30] implementation.
- 4) **XGBoost** [5]: 100 trees with L1/L2 regularization. Default hyperparameters.
- 5) **MLP** [32]: Two hidden layers (64 and 32 units), ReLU activation, early stopping. Included as a neural network baseline to compare against tree-based ensembles on tabular data.

All implementations use Python 3.10 with scikit-learn version 1.3, XGBoost version 2.0, SHAP version 0.43, and statsmodels version 0.14 for econometric analyses. Hyperparameters are fixed at defaults (no grid search) to maximize reproducibility and minimize overfitting to validation folds.

D. EVALUATION PROTOCOLS

Following Stage 4 (Section III-D), we employ three evaluation protocols:

- **Hospital-level Group K-Fold** (primary): 5 folds with entire hospitals held out per fold, stratified by deficit proportion. Prevents within-hospital data leakage inherent in random CV on panel data.
- **Temporal validation**: Training on 2019–2022 ($n = 660$), testing on 2023 ($n = 165$). Evaluates robustness under distributional shift (deficit rate changes from ~34% to 69.1%).
- **Stratified random 5-Fold** (reference): Standard CV with random observation-level splits. Included for comparison; may overestimate performance due to within-hospital leakage.

Performance metrics: AUC-ROC [33], [34] (primary), F1 score, and accuracy. All reported as means across folds with seed = 42.

E. COUNTERFACTUAL ANALYSIS

To demonstrate the framework's utility beyond prediction, we construct subsidy-adjusted distress trajectories. For each hospital i in year $t \in \{2020, \dots, 2023\}$:

$$NI_{i,t}^{\text{adj}} = NI_{i,t} - (DR_{i,t} - DR_{i,2019}) \quad (1)$$

where $NI_{i,t}$ is observed net income and $DR_{i,t}$ is observed donation revenue. Hospital i is classified as in counterfactual distress if $NI_{i,t}^{\text{adj}} < 0$. This analysis uses each hospital as its own control, isolating the effect of temporary external transfers on observed distress rates.

F. ECONOMETRIC VALIDATION

We validate the ML findings using panel fixed effects regression:

$$Y_{it} = \alpha_i + \mathbf{X}_{it}'\boldsymbol{\beta} + \boldsymbol{\gamma}'\mathbf{D}_t + \epsilon_{it} \quad (2)$$

where Y_{it} is the deficit indicator, α_i captures entity fixed effects (absorbing all time-invariant characteristics), $\mathbf{X}_{it}$ contains time-varying financial controls, and $\mathbf{D}_t$ are period dummies. Standard errors are clustered at the hospital level [35]. We estimate specifications with and without mediating financial controls to distinguish total period effects from direct period effects [31].

As a supplementary analysis, we employ an interrupted time series (ITS) design [36]:

$$Y_{it} = \alpha_i + \beta_1 \text{time}_t + \beta_2 \text{COVID_sub}_t + \beta_3 \text{Post_sub}_t + \mathbf{X}_{it}'\boldsymbol{\delta} + \epsilon_{it} \quad (3)$$

We note that only a single pre-intervention year (2019) is available, limiting the reliability of trend estimates; the ITS results are therefore exploratory.

V. RESULTS

A. MODEL PERFORMANCE COMPARISON

Table 5 presents classifier performance under all three evaluation protocols.

Under the primary Group K-Fold protocol, XGBoost achieves the highest AUC (0.842), closely followed by Gradient Boosting (0.840) and Random Forest (0.831). Logistic Regression (0.776) underperforms, indicating the presence of nonlinear feature interactions that linear models cannot capture. Notably, the modified Altman Z'-score achieves only AUC = 0.680. While the Z-score represents a 1990s-era static formula and is not a strictly fair baseline for modern ML, the more relevant baseline is Logistic Regression trained on the same data (AUC = 0.776). The 0.066 AUC lift provided by XGBoost (0.842) over Logistic Regression demonstrates that while linear accounting relationships explain much of the variance, capturing nonlinear feature interactions (such as institutional type interacting

TABLE 5. Model performance under three evaluation protocols.

Protocol	Model	AUC	F1	Acc
Group K-Fold [†]	XGBoost	0.842	0.708	0.771
	Gradient Boosting	0.840	0.704	0.771
	Random Forest	0.831	0.676	0.750
	Logistic Regression	0.776	0.636	0.732
	MLP (64,32)	0.772	0.583	0.717
	Altman Z''-score	0.680	0.601	0.605
Temporal (2023 test) [§]	Gradient Boosting	0.780	0.659	0.636
	Year-only baseline	0.659	—	—
Random 5-Fold [‡]	Gradient Boosting	0.857	0.726	0.787
	Random Forest	0.860	0.718	0.780
	Logistic Regression	0.775	0.617	0.719

[†] **Primary evaluation.** Hospital-level hold-out prevents within-entity leakage.

[‡] Reference. May overestimate due to within-hospital information sharing.

[§] Temporal holdout evaluates robustness under distributional shift.

^{||} Traditional credit scoring baseline. Z'' < 1.1 classified as distress.

All models: default hyperparameters, seed = 42. Bold = best per protocol.

with cost ratios) is critical for accurate detection. To assess whether the AUC differences among the top models are statistically significant, we apply the DeLong test [37] for pairwise comparison of correlated ROC curves on pooled out-of-fold predictions. Pairwise differences among the three ensemble models are not statistically significant: XGBoost vs. Gradient Boosting ($z = 0.848$, $p = 0.396$), XGBoost vs. Random Forest ($z = -0.603$, $p = 0.546$), and Random Forest vs. Gradient Boosting ($z = -0.107$, $p = 0.915$). Comparing ensemble models against Logistic Regression, only XGBoost achieves a statistically significant advantage ($z = 2.193$, $p = 0.028$), while Gradient Boosting ($z = 1.869$, $p = 0.062$) and Random Forest ($z = 1.770$, $p = 0.077$) fall short of the 5% threshold. These results indicate that the three ensemble methods perform comparably and that the ensemble advantage over the linear baseline, while consistent in direction, is statistically confirmed only for XGBoost. A multi-layer perceptron (MLP) with two hidden layers (64 and 32 units) achieves AUC = 0.772 under Group K-Fold, underperforming all tree-based ensembles. This is consistent with established findings that gradient-boosted trees outperform neural networks on moderate-sized tabular datasets [32].

The performance gap between Group K-Fold and random CV quantifies the degree of within-hospital data leakage. For Random Forest, AUC drops from 0.860 (random CV) to 0.831 (Group K-Fold): $\Delta\text{AUC} = -0.029$ (3.4%). This gap underscores the importance of entity-level holdout for unbiased evaluation in panel data settings.

Temporal validation (AUC = 0.780) reveals further performance degradation under distributional shift. The 2023 test set has a deficit rate of 69.1%, compared to ~34% in the training period (2019–2022), creating a substantial covariate

shift. Despite this, the model retains useful discriminative power well above that of a naïve year-only baseline (AUC = 0.659), demonstrating that the learned cost-structure features generalize beyond the training distribution.

B. ROBUSTNESS ANALYSES

To address concerns about feature–target leakage, parameter sensitivity, and temporal generalization, we conduct four additional analyses (XGBoost) using a stratified hospital-grouped five-fold protocol. Under this protocol the full 11-feature model attains AUC = 0.827—marginally below the 0.842 reported in Table 5 under the primary Group K-Fold split, a difference attributable to fold construction; we therefore emphasize *relative* changes, which are internally consistent within this protocol.

1) FEATURE ABLATION (SIGNAL BEYOND ACCOUNTING IDENTITIES)

Because the deficit label is derived from net income, some cost ratios are correlated with the target; we test whether the model captures signal beyond such accounting relationships. Removing the two features previously flagged as potentially endogenous (`subsidy_ratio` and `year`) reduces AUC only marginally, to 0.794. A stricter variant that additionally removes all three cost-structure ratios still achieves AUC = 0.651—well above the 0.5 chance level—using only structural and institutional features (debt ratio, size, hospital type, and ownership). These results indicate that the model learns substantial predictive signal that is not a mechanical restatement of the distress definition.

2) FORWARD (NEXT-YEAR) PREDICTION

To eliminate any contemporaneous mechanical link between features and label, we predict year- $(t+1)$ deficit from year- t features (665 hospital-year pairs). The model achieves AUC = 0.772, confirming genuine forward-looking predictive value rather than tautological in-year association.

3) ROLLING-ORIGIN TEMPORAL VALIDATION

Beyond the single 2023 holdout, we evaluate an expanding-window scheme (train 2019, test 2020; train 2019–2020, test 2021; and so forth). Per-split AUCs are 0.731, 0.850, 0.808, and 0.809 (mean 0.800), demonstrating stable temporal generalization despite year-to-year shifts in deficit prevalence (26–69%).

4) HYPERPARAMETER SENSITIVITY

Although our main results use default hyperparameters for reproducibility, we verify that this choice neither understates nor inflates performance. A randomized search (25 configurations) over XGBoost hyperparameters under the same protocol yields a best AUC of 0.834, essentially matching the default-parameter result (0.827). Performance is therefore robust to tuning and not an artifact of parameter selection.

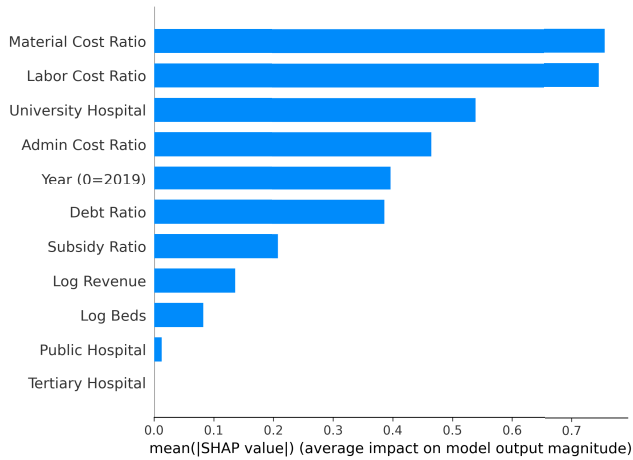


FIGURE 1. Mean absolute SHAP values for the Gradient Boosting model. Cost structure ratios (material, labor) dominate, followed by institutional type (university) and administrative cost ratio.

C. SHAP-BASED FEATURE IMPORTANCE

We select the Gradient Boosting model for SHAP analysis because its sequential boosting structure captures feature interactions more transparently than Random Forest's parallel bagging, yielding more interpretable SHAP interaction effects [16]. Although XGBoost achieves the highest AUC (0.842), the three ensemble models perform comparably (DeLong $p > 0.39$ for all pairwise comparisons), and SHAP feature-importance rankings are highly consistent across all three, as confirmed by Spearman rank correlation: RF-GB $\rho = 0.936$ ($p < 0.001$), RF-XGB $\rho = 0.964$ ($p < 0.001$), and GB-XGB $\rho = 0.982$ ($p < 0.001$).

Fig. 1 presents mean absolute SHAP values for the Gradient Boosting model, and Fig. 2 shows the SHAP beeswarm plot.

1) GLOBAL FEATURE IMPORTANCE

Material cost ratio ranks first (mean |SHAP| = 0.777), followed by labor cost ratio (0.729), university hospital indicator (0.537), administrative cost ratio (0.479), year (0.377), and debt ratio (0.359). Subsidy ratio ranks seventh (0.209) despite having the strongest directional protective effect.

The dominance of cost structure ratios (which collectively account for the top two and fourth-ranked features) indicates that organizational distress in this domain is fundamentally driven by cost-side dynamics rather than revenue shortfalls or financial leverage.

2) DIRECTIONAL EFFECTS

The beeswarm plot (Fig. 2) reveals monotonic directional patterns:

- **Material cost ratio:** High values (red) consistently push predictions toward distress, confirming that material cost burden is a primary risk factor.
- **Labor cost ratio:** Same directional pattern, with comparable effect magnitude.

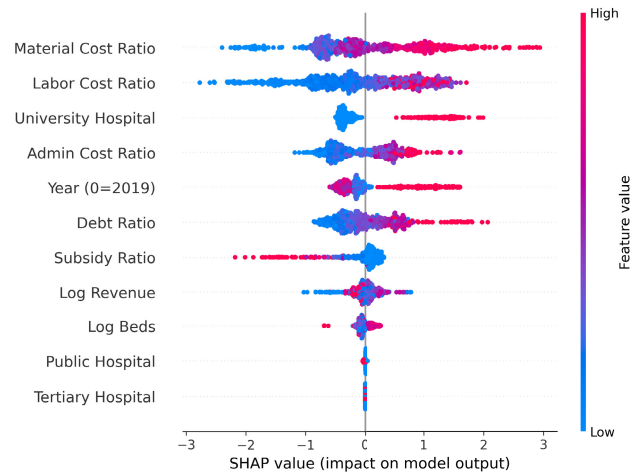


FIGURE 2. SHAP beeswarm plot. Each point is a hospital-year observation; red (blue) = high (low) feature value. Positive SHAP pushes prediction toward distress. Material and labor cost ratios show clear monotonic risk effects; subsidy ratio shows strong local protection but limited global importance.

- **University indicator:** University status provides a constant positive SHAP contribution regardless of other features—a “structural risk premium” reflecting institutional cost burdens.
- **Subsidy ratio:** High values push predictions strongly away from distress, confirming the protective effect of external transfers. However, most observations have near-zero subsidy ratios (median = 0.1%, IQR 0.0–1.8%), limiting global importance despite strong local effects.
- **Year:** Later years (red) increase distress probability, capturing the temporal deterioration trend.

3) RECONCILING SHAP AND LOGISTIC REGRESSION

A pooled logistic regression with clustered standard errors identifies the subsidy ratio as the strongest single predictor (marginal effect -1.288 , $p = 0.002$), despite its 7th-rank SHAP position. This apparent contradiction is resolved through a variance-vs-effect size analysis:

- **Logistic regression coefficients** measure per-unit directional effects; subsidy ratio has an extremely strong per-unit protective effect.
- **SHAP importance** reflects both effect size and variance in the feature. Subsidy ratio has very low variance (std = 9.7%), so its high per-unit effect is rarely exercised, yielding low global SHAP importance.
- Cost ratios have both substantial effects and high variance across the panel, producing large SHAP values.

This analysis demonstrates the importance of reporting both SHAP and linear model results: SHAP identifies which features *actually drive predictions* across the observed data, while logistic regression coefficients identify which features would *most change predictions if altered*—both are valuable in different decision contexts.

TABLE 6. Observed vs. Counterfactual (Subsidy-adjusted) distress rates.

	2020	2021	2022	2023
Donation rev. (trillion KRW)	—	2.159	—	0.530
2019 baseline (trillion KRW)	0.203			
Observed distress (%)	46.7	26.1	29.7	69.1
Counterfactual distress (%)	81.2	83.6	73.9	86.7
Masking effect (pp)	34.5	57.5	44.2	17.6

Counterfactual: each entity’s external transfer replaced with 2019 level (Eq. 1). pp = percentage points. 2019 observed distress: 35.2%. This analysis assumes no behavioral response to subsidy absence—a strong simplifying assumption. In practice, hospitals receiving subsidies may have expanded staffing, delayed cost-cutting, or undertaken capital investments that they would not have pursued absent external support. The counterfactual distress rates should therefore be interpreted as upper bounds rather than precise estimates of latent distress.

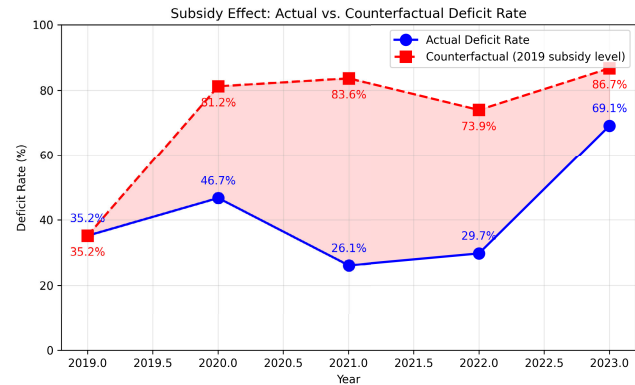


FIGURE 3. Observed vs. counterfactual distress rates (2019–2023). The counterfactual scenario replaces external transfers with pre-shock baselines, revealing latent distress masked by temporary subsidies. The 57.5 pp gap in 2021 indicates that the apparent “recovery” was entirely transfer-driven.

D. COUNTERFACTUAL DISTRESS ANALYSIS

Table 6 and Fig. 3 present the counterfactual (subsidy-adjusted) distress trajectories.

The counterfactual analysis reveals a maximum masking effect of 57.5 percentage points in 2021: the observed distress rate of 26.1% concealed a latent rate of 83.6%. Counterfactual distress rates remained stable at 73.9–86.7% across all post-2019 years, indicating continuous underlying deterioration. The variation in observed rates (26.1–69.1%) was driven almost entirely by variation in external transfer levels rather than by changes in operational performance. Even in 2023, when transfers had substantially declined, a 17.6 pp gap persisted, indicating that residual transfers continued to suppress the reported distress rate.

This analysis demonstrates a key capability of the proposed framework: by combining SHAP-identified features with domain-specific counterfactual construction, analysts can distinguish between *observed* and *structural* distress—a distinction that is invisible in standard classification metrics but critical for policy and management decisions.

TABLE 7. Distress rates by institutional category (2019 vs. 2023).

Subgroup	2019 (%)	2023 (%)	Test
<i>Hospital Type</i>			
Tertiary (<i>n</i> = 27)	66.7	59.3	χ^2 <i>p</i> -value
General (<i>n</i> = 138)	29.0	71.0	
	0.0004	0.3265	
<i>Establishment Type</i>			
Medical corp. (<i>n</i> = 74)	14.9	54.1	Kruskal–Wallis <i>H</i>
Public (<i>n</i> = 42)	47.6	85.7	
University (<i>n</i> = 35)	71.4	80.0	
<i>Net Income Change (2019→2023)</i>			
	17.26	<i>p</i> = 0.0002	

χ^2 tests compare distress rates across types within each year.

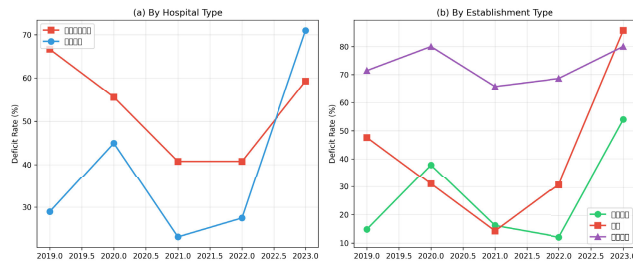


FIGURE 4. Distress rates by hospital type (left) and establishment type (right), 2019–2023. A convergence pattern emerges: previously heterogeneous groups become statistically indistinguishable by 2023, indicating a systemic rather than institution-specific crisis.

E. SUBGROUP ANALYSIS

The SHAP analysis identified institutional type as a key feature (university indicator ranked 3rd). Table 7 and Fig. 4 present subgroup distress rates.

A striking convergence pattern emerges: in 2019, hospital type was a strong distress predictor ($\chi^2 p = 0.0004$), but by 2023, the same test is non-significant ($p = 0.3265$). From a machine learning perspective, this means that categorical features that are predictive early in the panel lose discriminative power as the distress mechanism becomes systemic. This finding motivates the inclusion of temporal features and cost ratios—which retain predictive power throughout the observation period—in preference to static institutional categories.

The Kruskal–Wallis test ($H = 17.26$, $p = 0.0002$) confirms significant heterogeneity across establishment types in the degree of financial deterioration, validating the SHAP finding that institutional type captures meaningful variance in distress risk.

F. ECONOMETRIC VALIDATION

Table 8 presents panel fixed effects regression results, providing econometric triangulation of the ML findings.

The econometric results validate the SHAP analysis on multiple dimensions:

- **Material cost ratio:** SHAP rank #1, econometrically significant (+1.680, $p = 0.010$ in Model 1; +3.300, $p < 0.001$ in Model 3b).
- **Subsidy ratio:** Strong negative coefficient (−0.191, $p = 0.025$), confirming the protective effect identified by

TABLE 8. Panel fixed effects regression results (Econometric validation).

	Model 1 Deficit ~ Year	Model 2 NI Margin ~ Year	Model 3a Deficit ~ Period (total)	Model 3b Deficit ~ Period (direct)
<i>Year/Period Dummies</i>				
year_2020	+0.0955* (0.037)			
year_2021	−0.0536 (0.230)	+0.0545* (0.017)		
year_2022	−0.0340 (0.485)			
year_2023	+0.3288*** (<0.001)	−0.0096 (0.782)		
COVID+Subsidy (2020–22)			+0.0026 (0.945)	+0.0174 (0.663)
Post-Subsidy (2023)			+0.2894*** (<0.001)	+0.1361** (0.003)
<i>Financial Controls (SHAP-identified features)</i>				
material_ratio	+1.6796** (0.010)	−2.7224*** (<0.001)	—	+3.3000*** (<0.001)
subsidy_ratio	−0.1913* (0.025)	+0.7573*** (<0.001)	—	−0.2291* (0.029)
labor_ratio	+0.1356 (0.543)			
debt_ratio	+0.0130 (0.157)			
log_revenue	−0.1273 (0.140)			
Hospital FE	Yes	Yes	Yes	Yes
Clustered SE	Yes	Yes	Yes	Yes
R ² within	0.2078	0.4722	—	—
Observations	825	825	825	825

p-values in parentheses. ****p* < 0.001; ***p* < 0.01; **p* < 0.05. Models 3a/3b: without/with mediating financial controls (subsidy_ratio, material_ratio). The coefficient reduction from 3a to 3b (+0.2894 → +0.1361) indicates that approximately half of the post-subsidy effect operates through contemporaneous financial ratios.

SHAP despite its low global ranking—consistent with the variance-vs-effect size explanation.

- **Post-subsidy structural break:** The total effect (Model 3a: +0.289, *p* < 0.001) is absorbed when mediating financial controls are included (Model 3b: COVID coefficient *p* = 0.663), indicating that the distress mechanism operates *through* the cost and subsidy channels identified by SHAP.

The convergence between ML feature importance (SHAP) and econometric significance (panel FE) provides strong triangulated evidence that cost structure features are the fundamental drivers of organizational distress in this domain.

1) INTERRUPTED TIME SERIES (EXPLORATORY)

Table 9 presents the ITS results. The ITS analysis yields a post-subsidy level shift of +0.439 (*p* < 0.001), suggesting an abrupt structural break. An extended specification separating level and trend effects finds a pre-existing trend of −0.037 (*p* = 0.007) and a post-subsidy level shift of +0.439 (*p* < 0.001), suggesting sudden reversal rather than gradual deterioration. These results should be interpreted cautiously given the single pre-intervention year; they are presented as supplementary evidence consistent with the primary fixed effects findings.

VI. DISCUSSION

A. TECHNICAL INSIGHTS

1) ENTITY-LEVEL CV IS ESSENTIAL FOR PANEL DATA

The 0.029 AUC gap between random CV (0.860) and Group K-Fold (0.831) for Random Forest demonstrates that standard cross-validation overestimates performance on panel data. This gap (3.4%) can be even larger in settings with more observations per entity or with stronger entity-level effects. We recommend Group K-Fold as the default evaluation

protocol for ML models applied to panel data, reporting random CV only as a reference.

2) TEMPORAL ROBUSTNESS UNDER DISTRIBUTIONAL SHIFT

The further degradation from Group K-Fold (0.842) to temporal validation (0.780) quantifies the cost of distributional shift. The 2023 test year represents a regime change (deficit rate nearly doubling), yet the model retains useful discriminative power. This suggests that the learned cost-structure features capture fundamental distress mechanisms that transfer across regimes, rather than fitting to regime-specific patterns. For operational deployment, we recommend periodic model retraining as the distress landscape evolves.

3) CHOICE OF SHAP AND ITS LIMITATIONS

We adopt SHAP rather than alternative post-hoc explainers such as LIME [23] for three reasons. First, SHAP attributions are grounded in the Shapley value from cooperative game theory [24], which uniquely satisfies the efficiency, symmetry, and additivity axioms and therefore yields internally consistent, fairly allocated feature contributions [15]. Second, for the tree-ensemble models that perform best in our experiments, TreeExplainer computes exact Shapley values efficiently [16], whereas LIME relies on local surrogate approximations whose stability has been questioned in comparative financial-risk studies [26]. Third, SHAP provides a unified global and local view that our pipeline exploits for both population-level importance and case-level explanation. SHAP is nonetheless not without limitations. Its attributions assume an additive feature-contribution structure and can be difficult to interpret under strong feature interactions; correlated or compositional inputs—as with our cost-ratio shares (discussed below)—split credit in ways that destabilize

TABLE 9. Interrupted time series results (Exploratory).

	Model A No controls	Model B With controls	Model C Trend + Level
<i>Period Effects</i>			
COVID+Subsidy	−0.0101 (0.778)	−0.0166 (0.647)	
Post-Subsidy	+0.3394*** (<0.001)	+0.2894*** (<0.001)	
<i>Trend Components</i>			
time			−0.0370** (0.007)
post_level			+0.4394*** (<0.001)
<i>Financial Controls</i>			
material_ratio		+1.9680** (0.002)	
subsidy_ratio		−0.2146* (0.022)	
Hospital FE	Yes	Yes	Yes
Clustered SE	Yes	Yes	Yes
R ² within	0.1420	0.1828	—
Observations	825	825	825

p-values in parentheses. ****p* < 0.001; ***p* < 0.01; **p* < 0.05. Single pre-intervention year (2019) limits trend estimate reliability.

individual attributions even when group-level importance is robust [14]. Moreover, SHAP explains the model's learned associations rather than causal mechanisms [13]: a high attribution signals predictive reliance, not that intervening on the feature would change the outcome. For these reasons we report SHAP alongside linear-model coefficients and treat the two as complementary rather than interchangeable, as detailed next.

4) VARIANCE-vs-EFFECT SIZE IN SHAP INTERPRETATION

The divergence between SHAP rankings and logistic regression coefficients highlights a methodological subtlety in SHAP interpretation. Features with strong directional effects but low population variance (e.g., subsidy ratio) appear important in linear models but rank low in SHAP because their effects are rarely “activated” across the observed data distribution. Conversely, features with moderate effects but high variance (e.g., cost ratios) dominate SHAP because they explain a greater share of the variance in actual predictions. We recommend reporting both SHAP and linear model results for robust interpretation: SHAP for understanding *what drives predictions in practice*, and linear coefficients for understanding *what would change predictions if manipulated*.

5) COMPOSITIONAL FEATURE INSTABILITY

The three cost ratio features (labor, material, and administrative) are constrained to approximate shares of total cost. This creates collinearity that can produce instability in individual SHAP attributions across model refits. However, the aggregate finding—that cost ratios collectively dominate prediction—is robust. For deployment, we recommend tracking cost-ratio *groups* rather than individual ratios when communicating SHAP results to non-technical stakeholders.

6) LIMITS OF ECONOMETRIC CAUSAL INFERENCE

While panel fixed effects absorb all time-invariant hospital characteristics, the post-subsidy coefficient may capture the

effects of other concurrent changes (inflation, labor market tightening, regulatory shifts) beyond subsidy withdrawal alone. We interpret the econometric results as providing correlational triangulation of the ML findings rather than definitive causal evidence.

7) ON THE CHOICE OF BASELINES

We include a modified Altman Z''-score [21] as a baseline to benchmark ML performance against traditional credit scoring. The Z''-score, designed for non-manufacturing and private firms, uses the formula $Z'' = 6.56 X_1 + 3.26 X_2 + 6.72 X_3 + 1.05 X_4$, where X_1 = working capital/total assets, X_2 = retained earnings/total assets, X_3 = EBIT/total assets, and X_4 = book value of equity/total liabilities. For hospitals, we substitute medical operating income for EBIT and accumulated surplus/deficit for retained earnings. The Z-score achieves AUC = 0.680 under Group K-Fold cross-validation—substantially below all ML models (XGBoost: 0.842)—validating the need for domain-adaptive ML approaches over traditional credit scoring for non-market organizations.

B. FRAMEWORK GENERALIZABILITY

The proposed framework is designed with generalization beyond the hospital domain in mind, but we emphasize that the present study validates it on a single domain; broader applicability therefore remains a hypothesis to be confirmed empirically (see Section VI-G). While the individual components (ensemble learning, SHAP, and cross-validation) are well-established, the systematic integration of entity-level evaluation, variance-effect reconciliation, and counterfactual analysis within a single pipeline addresses gaps in how these tools are typically applied to organizational finance. In principle, the four stages—domain-adaptive feature engineering, multi-model comparison, SHAP interpretation, and domain-specific validation—are intended to apply to other organizational settings characterized by:

- 1) **Structured financial disclosures:** Mandatory or voluntary reporting of balance sheet and income statement data. Examples include municipal governments (CAFR reports), universities (IPEDS financial data in the US), and regulated utilities (FERC filings).
- 2) **Panel structure:** Repeated observations of the same entities over time, enabling entity-level cross-validation and fixed effects econometric validation.
- 3) **Domain-specific features:** Institutional characteristics, regulatory constraints, or external transfer mechanisms that require domain knowledge for feature engineering.
- 4) **Non-market distress dynamics:** Organizations where distress manifests as progressive margin compression and operating deficits rather than stock price decline or debt default.

By design, most domain expertise is concentrated in Stage 1 (feature engineering), while Stages 2–4 are intended to transfer with limited modification. We stress, however, that this transferability is an architectural design goal rather than an empirically demonstrated result; confirming it would require instantiating the framework on the additional domains noted above.

C. DOMAIN-SPECIFIC INSIGHTS: HOSPITAL FINANCIAL DISTRESS

Although the primary contribution of this work is methodological, the hospital domain application yields findings of independent interest.

1) REVENUE–COST SCISSORS AS A DISTRESS MECHANISM

The fundamental distress mechanism in the hospital domain is a “scissors effect” in which cost growth (+29.7%) systematically outpaces revenue growth (+23.1%) due to structural asymmetry: costs are market-driven (labor shortages, input inflation) while revenues are policy-constrained (regulated fee schedules). This mechanism is visible in both the SHAP analysis (cost ratios dominate) and the econometric results (material cost ratio is significant across all specifications).

Labor costs account for 48.1% of the total cost increase, reflecting minimum wage increases, mandatory staffing ratio requirements, and pandemic-era hiring commitments that proved difficult to reverse. The labor cost-to-revenue ratio increased from 45.7% to 48.5%, representing a 2.8 percentage point structural margin compression.

2) EXTERNAL TRANSFER MASKING

The counterfactual analysis demonstrates a phenomenon we term “transfer masking”: temporary external subsidies can suppress observed distress rates while underlying structural deterioration continues unchecked. The maximum masking effect (57.5 pp in 2021) created the appearance of financial recovery that did not reflect operational reality, potentially delaying structural interventions [22]. This finding has implications beyond healthcare: any organizational setting

TABLE 10. Sensitivity to distress definition (GB, Group K-fold).

Definition	Deficit (%)	AUC	<i>n</i>
Net Income < 0 (baseline)	41.4	0.836	829*
Operating Deficit	57.9	0.952	829*
Two Consecutive Years	22.7	0.738	664

* Sensitivity analysis used slightly different preprocessing, yielding 829 observations vs. 825 in the main analysis.

Operating Deficit = medical revenue – medical cost < 0. Two Consecutive Years requires net income < 0 in both year t and $t - 1$, restricting the sample to 2020–2023 ($n = 664$).

in which external transfers (government bailouts, cross-subsidies, endowment drawdowns) coexist with structural cost pressures is susceptible to transfer masking.

3) INSTITUTIONAL CONVERGENCE

The loss of discriminative power for the hospital type feature ($\chi^2 p: 0.0004 \rightarrow 0.3265$) indicates that the distress mechanism became systemic by 2023, affecting organizations across all categories. From a modeling perspective, this suggests that static institutional features lose predictive value during systemic crises, and that dynamic cost-structure features should receive greater weight in model design for crisis-period prediction.

D. SENSITIVITY TO DISTRESS DEFINITION

The baseline analysis defines financial distress as net income < 0. To evaluate whether the framework generalizes across alternative distress definitions, we retrain the Gradient Boosting model under Group K-Fold cross-validation using two additional definitions: (1) operating deficit (medical revenue – medical cost < 0), and (2) two consecutive years of net loss. Table 10 summarizes the results.

The framework generalizes across distress definitions, though predictive performance varies substantially. Operating deficit is the most predictable target (AUC = 0.952), reflecting the near-deterministic relationship between cost-structure ratios and the medical operating margin—features that directly decompose the target variable. The baseline net income definition (AUC = 0.836) introduces additional noise from non-operating items (subsidies, financial income/expenses), moderately reducing predictability. Consecutive-year distress is the hardest to predict (AUC = 0.738) because it introduces temporal dependencies: whether a hospital experienced a loss in the *prior* year is not directly captured by the current year’s financial ratios. These results confirm that the framework is robust to the operationalization of distress, while highlighting the importance of matching the distress definition to the analytical objective.

E. ALGEBRAIC RELATIONSHIP BETWEEN FEATURES AND TARGET

We acknowledge that cost structure ratios—the dominant predictive features—bear a strong algebraic relationship to

the detection target (net income deficit). Because net income is mathematically derived from revenue and costs, these features raise concerns about endogeneity. However, the ML model's utility lies not in relearning the $Revenue - Cost < 0$ accounting identity, but in identifying the specific, nonlinear thresholds and interactive conditions (e.g., how university status interacts with labor ratios) under which distress materializes across different organizational structures. Furthermore, the counterfactual analysis replacing external transfers with pre-shock baselines assumes no behavioral response from hospitals. In reality, hospitals facing severe deficits would likely implement cost-cutting measures or reduce services. The 83.6% counterfactual deficit rate should therefore be interpreted as a theoretical upper bound of distress exposure, not a precise forecast of alternative historical events.

F. RELATIONSHIP TO COMPANION UNSUPERVISED ANALYSIS

A companion unsupervised analysis of hospital financial patterns is reported separately (Kim and Sun, submitted). The two studies address different analytical needs: the supervised approach developed here optimizes prediction of a labeled distress outcome, whereas the unsupervised approach identifies structurally unusual patterns without predefined categories. Treating them separately preserves the distinct methodological contribution of each.

G. LIMITATIONS

Several limitations should be acknowledged. First, our panel includes only tertiary and general hospitals; long-term care and psychiatric hospitals are excluded, limiting generalizability within the healthcare domain. Second, the subsidy proxy (donation revenue) may include non-governmental donations and may miss some government support recorded through other accounting channels. Third, the ITS analysis has only a single pre-intervention year (2019), whereas Bernal et al. [36] recommend eight; the ITS results are therefore exploratory. Fourth, while panel fixed effects strengthen causal interpretation relative to cross-sectional analysis, we do not establish definitive causal relationships [38]. Fifth, the balanced panel requirement excludes hospitals that entered or exited during 2019–2023, introducing survivorship bias that may underestimate distress prevalence among the broader hospital population. Sixth, the `year` and `subsidy_ratio` features raise endogeneity concerns; our ablation analysis (Section V-B) shows the model retains $AUC = 0.794$ when these features are removed and $AUC = 0.651$ when all cost ratios are additionally removed, indicating predictive signal beyond accounting identities, and a next-year prediction setting yields $AUC = 0.772$. Seventh, hyperparameters are fixed at defaults; a randomized hyperparameter search (Section V-B) yields essentially identical performance ($AUC = 0.834$ vs. 0.827), confirming robustness to tuning. Eighth, the framework has been validated on a single domain; generalization to other

organizational types (municipalities, universities) requires empirical confirmation.

VII. CONCLUSION

We have presented a systematic explainable machine learning framework for organizational financial distress detection, comprising domain-adaptive feature engineering, multi-model comparison, SHAP-based interpretation, and domain-specific validation. Instantiated on Korean hospital panel data (165 hospitals, 825 observations, 2019–2023), the framework achieves $AUC = 0.842$ under entity-level Group K-Fold cross-validation and $AUC = 0.780$ under temporal validation. SHAP analysis reveals that cost structure ratios (material: mean $|SHAP| = 0.777$; labor: 0.729) are the dominant predictive features, while external subsidies exert strong local protective effects that are invisible in global SHAP rankings due to their low variance. Counterfactual analysis exposes a 57.5 pp gap between observed and latent distress rates, demonstrating the framework's utility for scenario analysis. Panel fixed effects regression provides econometric confirmation of the ML-identified features and quantifies a post-subsidy structural break of 28.9 pp ($p < 0.001$).

The approach is designed to be adaptable to other organizational settings with structured financial disclosures and panel data. Future work should validate generalizability on municipal finance, university, and utility datasets; incorporate operational features (e.g., utilization rates and case mix); extend temporal coverage to enable more robust ITS analysis; and explore deep learning approaches with attention-based interpretability for richer feature spaces.

Analysis code and data documentation are available at <https://github.com/yjkim0123/hospital-financial-crisis>.

DATA AVAILABILITY STATEMENT

The hospital financial data used in this study are publicly available from the Korea Health Industry Development Institute (KHIDI) Hospital Financial Disclosure portal (<https://haspa.khidi.or.kr>). Analysis code is available at <https://github.com/yjkim0123/hospital-financial-crisis>

DECLARATION OF GENERATIVE AI USE

Generative AI tools were used for language editing and formatting assistance during manuscript preparation. All scientific content, data analysis, interpretation, and conclusions are solely the work of the authors, who take full responsibility for the accuracy and integrity of this paper.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

Yongjun Kim: Conceptualization, Methodology, Software, Data curation, Formal analysis, Visualization, Writing—original draft. Eun Jung Sun: Conceptualization, Methodology, Validation, Formal analysis, Writing—review and editing, Supervision.

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